Fibonacci sequences and linear algebra

Questions closer to the original prompts

Starting point. Let $F_{a,b}$ denote the sequence

$$(a, b, a+b, a+2b, 2a+3b, \ldots), \qquad u_0 = a, \quad u_1 = b, \quad u_{n+2} = u_{n+1} + u_n.$$

Work over the real numbers. The first two terms determine the whole sequence; our eventual goal is a direct formula for its nth term.

Q1. Can you give me an explicit term formula for any of these sequences?

Answer. Yes: $F_{0,0} = (0, 0, 0, \ldots)$ has $u_n = 0$ for every n. This simplest solution will be our zero vector.

Q2. Suppose I give you the formula for $F_{2,3}$. Which other formulas can you get from it?

Answer. We can scale every term. If $u_{n+2} = u_{n+1} + u_n$, then $(cu_{n+2}) = (cu_{n+1}) + (cu_n)$. Thus $cF_{2,3} = F_{2c,3c}$. For example, a formula for $F_{2,3}$ immediately gives one for $F_{4,6}$.

Q3. Starting from $F_{2,3}$, can scalar multiplication reach every sequence?

Answer. No. To get $F_{0,1}$, we would need $2c = 0$ and $3c = 1$ simultaneously. One sequence cannot generate every solution by scaling alone.

Q4. No? Then I'll supply a second sequence. Which should it be?

Answer. It should not be a scalar multiple of $F_{2,3}$: choosing $F_{4,6}$ would give us nothing new. A useful candidate is $F_{2,4}$, whose initial pair $(2, 4)$ is not proportional to $(2, 3)$.

Q5. Besides scaling, what operation could we try?

Answer. Addition! If $w_n = u_n + v_n$, then

$$w_{n+2} = u_{n+1} + u_n + v_{n+1} + v_n = w_{n+1} + w_n.$$

So we can form linear combinations, with $xF_{2,3} + yF_{2,4} = F_{2x+2y,3x+4y}$.

Q6. Take $F_{2,4}$. Is this enough? Yes? Well how do you get to $F_{0,1}$? And to $F_{\sqrt{2},1.5}$?

Answer. Yes. Solve $2x + 2y = a$, $3x + 4y = b$ to obtain $x = 2a - b$, $y = b - \frac{3}{2}a$. In particular,

$$\begin{aligned} F_{0,1} &= F_{2,4} - F_{2,3}, \\ F_{\sqrt{2},1.5} &= (2\sqrt{2} - 1.5)F_{2,3} + (1.5 - 1.5\sqrt{2})F_{2,4}. \end{aligned}$$

Matching the first two terms proves equality of the entire sequences.

The same prompts, now leading to a basis

Q7. What are the key properties of this pair?

Answer. Neither sequence is a scalar multiple of the other, and together they generate every solution. More precisely,

$$\boxed{F_{a,b} = (2a - b)F_{2,3} + \left(b - \tfrac{3}{2}a\right) F_{2,4}.}$$

The coefficients are unique. If $xF_{2,3} + yF_{2,4} = 0$, the equations $2x + 2y = 0$, $3x + 4y = 0$ force $x = y = 0$. These are the two properties of a *basis*: spanning and linear independence.

Q8. Could three sequences have these properties? What should independence mean for three?

Answer. For three sequences u, v, w, independence means that

$$\alpha u + \beta v + \gamma w = 0 \quad \Longrightarrow \quad \alpha = \beta = \gamma = 0.$$

It is not enough that no two are scalar multiples. For example, $F_{1,0}$, $F_{0,1}$, $F_{1,1}$ have that pairwise property, but

$$F_{1,0} + F_{0,1} - F_{1,1} = 0.$$

In fact, *any* three solutions are dependent. If their initial pairs are (a_i, b_i), choose a nonzero solution to

$$\lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 = 0, \qquad \lambda_1 b_1 + \lambda_2 b_2 + \lambda_3 b_3 = 0.$$

Such a solution exists because there are two homogeneous equations and three unknowns. The corresponding sequence combination starts with two zeros, so it is identically zero.

Q9. Does every independent pair let us obtain every sequence?

Answer. Yes. Let the pair be $F_{a,b}, F_{c,d}$. Independence is equivalent to $ad - bc \neq 0$. To obtain $F_{r,s}$, solve

$$ax + cy = r, \qquad bx + dy = s.$$

The determinant is nonzero, so there is a unique solution:

$$x = \frac{dr - cs}{ad - bc}, \qquad y = \frac{as - br}{ad - bc}.$$

Q10. Conversely, must a pair generating every sequence be independent?

Answer. Yes. A dependent pair spans at most the multiples of one nonzero sequence (or just the zero sequence). Its initial pairs therefore lie on a line through the origin and cannot include both $(1, 0)$ and $(0, 1)$. It cannot generate both $F_{1,0}$ and $F_{0,1}$.

Q11. What have we discovered? Does the zero sequence help us generate more?

Answer. The solution set V is a vector space under termwise operations. The map $F_{a,b} \mapsto (a, b)$ is a linear bijection $V \to \mathbb{R}^2$, so $\dim V = 2$. One vector cannot span it; three are dependent; and a pair spans exactly when it is independent. The zero vector adds nothing to a span, but belongs to every span, since multiplying any vector by 0 produces it.

Back to the goal: an actual term formula

Q12. But we still lack a term formula! What should our goal now be?

Answer. Find *two independent solutions with explicit formulas.* Since they will form a basis, we can obtain the formula for any $F_{a,b}$ by matching its first two entries. A basis is useful here only when its own terms are easy to compute.

Q13. Try arithmetic progressions. Do any useful examples work?

Answer. Suppose $u_n = A + nd$. The recurrence becomes

$$A + (n+2)d = 2A + (2n+1)d, \quad \text{or} \quad A + (n-1)d = 0 \quad (n \geq 0).$$

Taking $n = 1$ and $n = 0$ gives $A = d = 0$. Thus only the zero sequence works.

Q14. Now try geometric progressions. Which ratios work?

Answer. Try $u_n = q^n$, with $u_0 = 1$. At $n = 0$, the recurrence requires $q^2 = q + 1$. Conversely, this identity implies every later recurrence equation after multiplication by q^n. Solving the quadratic gives

$$\phi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$

We have therefore found the explicit solutions

$$g_\phi = (1, \phi, \phi^2, \ldots), \qquad g_\psi = (1, \psi, \psi^2, \ldots).$$

Q15. Can these explicit solutions supply a basis?

Answer. Yes. If one were a scalar multiple of the other, their equal first terms would force the scalar to be 1, contradicting $\phi \neq \psi$. Consequently every solution has the form

$$u_n = C\phi^n + D\psi^n.$$

For initial values (a, b), solve $C + D = a$, $C\phi + D\psi = b$:

$$C = \frac{b - a\psi}{\sqrt{5}}, \qquad D = \frac{a\phi - b}{\sqrt{5}}.$$

This gives an explicit formula for every sequence in the space.

Q16. Now obtain the formula for $F_{0,1}$.

Answer. Here $C + D = 0$ and $C\phi + D\psi = 1$. Since $\phi - \psi = \sqrt{5}$, we get $C = 1/\sqrt{5}$ and $D = -1/\sqrt{5}$. Thus

$$\boxed{F_n = \frac{\phi^n - \psi^n}{\sqrt{5}} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right].}$$

The expression satisfies the recurrence and starts with $0, 1$; those two facts prove it is the ordinary Fibonacci sequence.

Naming the ideas and extending the example

Q17. Match our informal discoveries with their formal linear-algebra names.

Answer. The two descriptions fit together as follows.

What we did with sequences	The formal idea
Collected all solutions and combined them termwise	Vector space
Used the all-zero solution	Zero vector
Scaled and added known solutions	Linear combination
Collected everything those operations produce	Span
Excluded nontrivial relations giving zero	Linear independence
Found two solutions generating everything uniquely	Basis; dimension two
Matched two initial values to find coefficients	Solving linear equations
Replaced an awkward basis by geometric solutions	Change of basis

Q18. What natural operation on sequences explains the geometric solutions?

Answer. Shift one place to the left:

$$S(u_0, u_1, u_2, \ldots) = (u_1, u_2, u_3, \ldots).$$

This is a linear map on V. A nonzero sequence is an eigenvector with eigenvalue q exactly when $Su = qu$, that is, $u_{n+1} = qu_n$ for all n. Equivalently, it is geometric: $u_n = u_0 q^n$, with $u_0 \neq 0$. For sequences in V, the possible ratios are precisely ϕ and ψ.

In initial-value coordinates, S sends (a, b) to $(b, a + b)$, so its matrix is

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

In the basis g_ϕ, g_ψ, its matrix is instead $\operatorname{diag}(\phi, \psi)$. The geometric basis makes the shift diagonal.

Q19. What changes for $u_{n+3} = u_{n+2} + u_{n+1} + u_n$?

Answer. Three initial entries determine the sequence, so its solution space is three-dimensional. The shift acts on initial triples by

$$(a, b, c) \longmapsto (b, c, a + b + c), \qquad M = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Trying $u_n = q^n$ now gives $q^3 = q^2 + q + 1$.

Attribution and changes. Based on Menny's Fibonacci teaching example, posted as a community-wiki MathOverflow answer by Theo Johnson-Freyd on 30 July 2010: mathoverflow.net/a/33960. This version follows the original order, examples and classroom voice more closely. Questions are adapted from the prompts; worked answers are supplied here. The source's typo in the proposed basis-finding goal is corrected: the desired sequences must be independent. Indexing begins at 0. Accessed 24 October 2026.